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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Chemical Equilibrium

JEE Main Chemistry · Previous Year Questions with Answer Key

131	111	2002 – 2026	English
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MCQ (Single Correct Answer) 81 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Consider the following reactions in which all the reactants and products are present in gaseous state

$2xy \rightleftharpoons x_2 + y_2 \;\; K_1 = 2.5 \times 10^5$

$xy + \frac{1}{2}z_2 \rightleftharpoons xyz \;\; K_2 = 5 \times 10^{-3}$

The value of K_3 for the equilibrium $\frac{1}{2}x_2 + \frac{1}{2}y_2 + \frac{1}{2}z_2 \rightleftharpoons xyz$ is :

(A) 2.5×10^{-3} (B) 2.5×10^{3} (C) 1.0×10^{-5} (D) 5×10^{-3}

Open solution

Q2. JEE Main 2026 (Online) 6th April Morning Shift

One mole each of He and $A(g)$ are taken in a 10 L closed flask and heated to 400 K to establish the following equilibrium.

$A(g) \rightleftharpoons B(g)$

K_c for this reaction at 400 K is 4.0 . The partial pressures (in atm) of He and B(g) are respectively (at equilibrium)

(Assume $He, A(g)$ and $B(g)$ behave as ideal gases)

(Given : $R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}$)

(A) 3.28, 2.624 (B) 2.624, 3.28 (C) 3.28, 0.656 (D) 0.656, 6.56

Open solution

Q3. JEE Main 2026 (Online) 5th April Evening Shift

The reaction $A(g) \rightleftharpoons B(g) + C(g)$ was initiated with the amount ' a ' of $A(g)$. At equilibrium it is found that the amount of $A(g)$ remaining is ($a - x$) at a total pressure of p .

The equilibrium constant Kp of the reaction can be calculated from the expression :

(A) $\frac{x^2}{a^2+x^2} \times p$ (B) $\frac{x^2}{a^2-x^2} \times p$

(C) $\frac{a+x^2}{x^2} \times p$ (D) $\frac{a^2-x^2}{x^2} \times p$

Open solution

Q4. JEE Main 2026 (Online) 4th April Morning Shift

At $T(K)$, the equilibrium constant of

$A_2(g) + B_2(g) \rightleftharpoons C(g)$ is 2.7×10^{-5}.

What is the equilibrium constant for

$\frac{1}{3}A_2(g) + \frac{1}{3}B_2(g) \rightleftharpoons \frac{1}{3}C(g)$ at the same temperature?

(A) $\left(2.7 \times 10^{-5}\right)^3$ (B) 6×10^{-2}

(C) $\sqrt{2.7 \times 10^{-5}}$ (D) 3×10^{-2}

Open solution

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Q5. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Observe the following equilibrium in a 1 L flask.

A(g) ⇌ B(g)

At T(K), the equilibrium concentrations of A and B are 0.5 M and 0.375 M respectively. 0.1 moles of A is added into the flask and heated to T(K) to establish the equilibrium again. The new equilibrium concentrations (in M) of A and B are respectively

(A) 0.742, 0.557. **(B)** 0.367, 0.275. **(C)** 0.53, 0.4. **(D)** 0.557, 0.418.

Q6. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Consider the following gaseous equilibrium in a closed container of volume ' V ' at $\mathrm{T(K)}$.

$\mathrm{P_2(g) + Q_2(g) \rightleftharpoons 2PQ(g)}$

2 moles each of $\mathrm{P_2(g)}$, $\mathrm{Q_2(g)}$ and $\mathrm{PQ(g)}$ are present at equilibrium. Now one mole each of ' $\mathrm{P_2}$ ' and ' $\mathrm{Q_2}$ ' are added to the equilibrium keeping the temperature at $\mathrm{T(K)}$. The number of moles of $\mathrm{P_2}, \mathrm{Q_2}$ and PQ at the new equilibrium, respectively, are

(A) $2.56, 1.62, 2.24$ **(B)** $2.67, 2.67, 2.67$ **(C)** 1.21, 2.24, 1.56 **(D)** $1.66, 1.66, 1.66$

Q7. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Consider the general reaction given below at 400 K

$x\,\mathrm{A(g)} \rightleftharpoons y\,\mathrm{B(g)}$.

The values of K_p and K_c are studied under the same condition of temperature but variation in x and y.

(i) $\mathrm{K_p = 85.87}$ and $\mathrm{K_c = 2.586}$ appropriate units

(ii) $\mathrm{K_p = 0.862}$ and $\mathrm{K_c = 28.62}$ appropriate units

The values of x and y in (i) and (ii) respectively are :

(A)

(i)	(ii)
1,2	2,1

(B)

(i)	(ii)
1,3	2,1

(C)

(i)	(ii)
3,1	3,1

(D)

(i)	(ii)
4,1	4,1

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Q8. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

In the following system, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ at equilibrium, upon addition of xenon gas at constant T & p , the concentration of

(A) $PCl_5, PCl_3 \& Cl_2$ remain constant

(B) PCl_3 will increase

(C) Cl_2 will decrease

(D) PCl_5 will increase

Q9. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Given below are two statements :

Statement I : A catalyst cannot alter the equilibrium constant (K_c) of the reaction, temperature remaining constant.

Statement II : A homogeneous catalyst can change the equilibrium composition of a system, temperature remaining constant.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q10. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Consider the following chemical equilibrium of the gas phase reaction at a constant temperature :
$A(g) \rightleftharpoons B(g) + C(g)$

If p being the total pressure, K_p is the pressure equilibrium constant and α is the degree of dissociation, then which of the following is true at equilibrium?

(A) If K_p value is extremely high compared to p, α becomes much less than unity

(B) When p increases α increases

(C) If p value is extremely high compared to $K_p, \alpha \approx 1$

(D) When p increases α decreases

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Q11. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Consider the equilibrium

$\mathrm{CO(g) + 3H_2(\,g) \rightleftharpoons CH_4(\,g) + H_2O(\,g)}$

If the pressure applied over the system increases by two fold at constant temperature then

(A) Concentration of reactants and products increases.

(B) Equilibrium will shift in forward direction.

(C) Equilibrium constant increases since concentration of products increases.

(D) Equilibrium constant remains unchanged as concentration of reactants and products remain same.

Choose the **correct** answer from the options given below :

(A) (A) and (B) only

(B) (A), (B) and (D) only

(C) (B) and (C) only

(D) (A), (B) and (C) only

Q12. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

At temperature T, compound $AB_{2(g)}$ dissociates as $AB_{2(g)} \rightleftharpoons AB_{(g)} + \frac{1}{2}B_{2(g)}$ having degree of dissociation x (small compared to unity). The correct expression for x in terms of K_p and p is:

(A) $\sqrt{K_p}$

(B) $\sqrt[3]{\frac{2K_p^2}{p}}$

(C) $\sqrt[3]{\frac{2K_p}{p}}$

(D) $\sqrt[4]{\frac{2K_p}{p}}$

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Q13. JEE Main 2025 (Online) 24th January Evening Shift

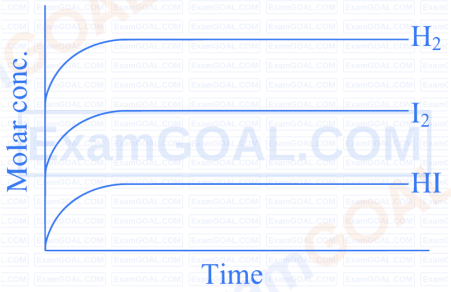
Open solution

For the reaction,

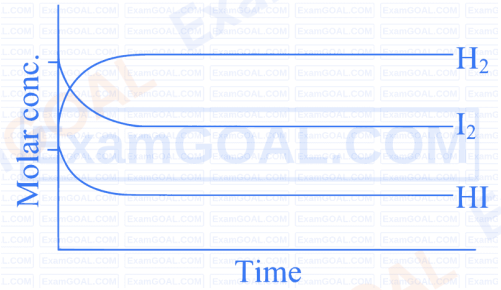
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

Attainment of equilibrium is predicted correctly by :

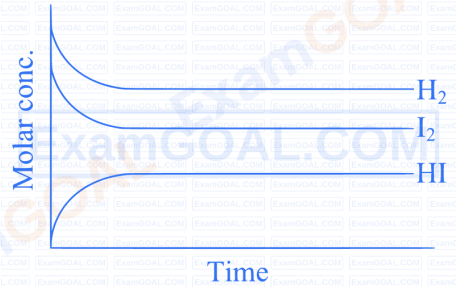
(A)



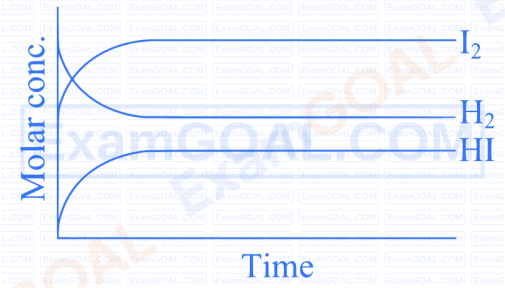
(B)



(C)



(D)



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Q14. JEE Main 2025 (Online) 23rd January Evening Shift

Consider the reaction

$X_2Y(g) \rightleftharpoons X_2(g) + \frac{1}{2}Y_2(g)$

The equation representing correct relationship between the degree of dissociation (x) of $X_2Y(g)$ with its equilibrium constant Kp is _____.

Assume x to be very very small.

(A) $x = \sqrt[3]{\frac{Kp}{p}}$ (B) $x = \sqrt[3]{\frac{Kp}{2p}}$

(C) $x = \sqrt[3]{\frac{2Kp^2}{p}}$ (D) $x = \sqrt[3]{\frac{2Kp}{p}}$

Open solution

Q15. JEE Main 2025 (Online) 22nd January Morning Shift

A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm . Some of CO_2 is converted into CO on addition of graphite. If total pressure at equilibrium is 0.8 atm , then K_p is :

(A) 0.18 atm (B) 0.3 atm (C) 3 atm (D) 1.8 atm

Open solution

Q16. JEE Main 2024 (Online) 8th April Morning Shift

For the given hypothetical reactions, the equilibrium constants are as follows :

$X \rightleftharpoons Y; K_1 = 1.0$

$Y \rightleftharpoons Z; K_2 = 2.0$

$Z \rightleftharpoons W; K_3 = 4.0$

The equilibrium constant for the reaction $X \rightleftharpoons W$ is

(A) 12.0 (B) 8.0 (C) 6.0 (D) 7.0

Open solution

Q17. JEE Main 2024 (Online) 6th April Evening Shift

The ratio $\frac{K_P}{K_C}$ for the reaction :

$CO_{(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons CO_{2(g)}$ is :

(A) 1 (B) $(RT)^{1/2}$ (C) RT (D) $\frac{1}{\sqrt{RT}}$

Open solution

Q18. JEE Main 2024 (Online) 6th April Morning Shift

At $-20^\circ C$ and $1\ atm$ pressure, a cylinder is filled with equal number of H_2, I_2 and HI molecules for the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, the K_p for the process is $x \times 10^{-1}$.

x = _____.

[Given : $R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}$]

(A) 2 (B) 1 (C) 10 (D) 0.01

Open solution

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Q19. JEE Main 2024 (Online) 5th April Evening Shift

Open solution

Given below are two statements :

Statement I : On passing $HCl_{(g)}$ through a saturated solution of $BaCl_2$, at room temperature white turbidity appears.

Statement II : When HCl gas is passed through a saturated solution of $NaCl$, sodium chloride is precipitated due to common ion effect.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Q20. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

The following reaction occurs in the Blast furnace where iron ore is reduced to iron metal

$Fe_2O_{3(s)} + 3CO_{(g)} \rightleftharpoons Fe_{(l)} + 3CO_{2(g)}$

Using the Le Chatelier's principle, predict which one of the following will not disturb the equilibrium.

(A) Addition of CO_2

(B) Removal of CO

(C) Addition of Fe_2O_3

(D) Removal of CO_2

Q21. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

The equilibrium constant for the reaction

$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$

is $K_c = 4.9 \times 10^{-2}$. The value of K_c for the reaction given below $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is :

(A) 49

(B) 416

(C) 41.6

(D) 4.9

Q22. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

$A_{(g)} \rightleftharpoons B_{(g)} + \frac{C}{2}(g)$ The correct relationship between K_P, α and equilibrium pressure P is

(A) $K_P = \frac{\alpha^{1/2}P^{3/2}}{(2+\alpha)^{3/2}}$

(B) $K_P = \frac{\alpha^{3/2}P^{1/2}}{(2+\alpha)^{1/2}(1-\alpha)}$

(C) $K_P = \frac{\alpha^{1/2}P^{1/2}}{(2+\alpha)^{3/2}}$

(D) $K_P = \frac{\alpha^{1/2}P^{1/2}}{(2+\alpha)^{1/2}}$

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Q23. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

For the given reaction, choose the correct expression of K_C from the following :-

$Fe^{3+}_{(aq)} + SCN^{-}_{(aq)} \rightleftharpoons (FeSCN)^{2+}_{(aq)}$

(A) $K_C = \frac{[Fe^{3+}][SCN^-]}{[FeSCN^{2+}]}$

(B) $K_C = \frac{[FeSCN^{2+}]}{[Fe^{3+}][SCN^-]}$

(C) $K_C = \frac{[FeSCN^{2+}]^2}{[Fe^{3+}][SCN^-]}$

(D) $K_C = \frac{[FeSCN^{2+}]}{[Fe^{3+}]^2[SCN^-]^2}$

Q24. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

For a concentrated solution of a weak electrolyte (K_{eq} = equilibrium constant) A_2B_3 of concentration 'c', the degree of dissociation 'α' is :

(A) $\left(\frac{K_{eq}}{25c^2}\right)^{\frac{1}{5}}$

(B) $\left(\frac{K_{eq}}{108c^4}\right)^{\frac{1}{5}}$

(C) $\left(\frac{K_{eq}}{5c^4}\right)^{\frac{1}{5}}$

(D) $\left(\frac{K_{eq}}{6c^5}\right)^{\frac{1}{5}}$

Q25. JEE Main 2022 (Online) 30th June Morning Shift

Open solution

The equilibrium constant for the reversible reaction

$2A(g) \rightleftharpoons 2B(g) + C(g)$ is K_1

$\frac{3}{2}A(g) \rightleftharpoons \frac{3}{2}B(g) + \frac{3}{4}C(g)$ is K_2.

K_1 and K_2 are related as :

(A) $K_1 = \sqrt{K_2}$

(B) $K_2 = \sqrt{K_1}$

(C) $K_2 = K_1^{3/4}$

(D) $K_1 = K_2^{3/4}$

Q26. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

4.0 moles of argon and 5.0 moles of PCl_5 are introduced into an evacuated flask of 100 litre capacity at 610 K. The system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The K_p for the reaction is :

[Given : R = 0.082 L atm K^{-1} mol^{-1}]

(A) 2.25

(B) 6.24

(C) 12.13

(D) 15.24

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Q27. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

For a reaction at equilibrium

$A(g) \rightleftharpoons B(g) + \frac{1}{2} C(g)$

the relation between dissociation constant (K), degree of dissociation (α) and equilibrium pressure (p) is given by :

(A) $K = \frac{\alpha^{\frac{1}{2}} p^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$

(B) $K = \frac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)}$

(C) $K = \frac{(\alpha p)^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$

(D) $K = \frac{(\alpha p)^{\frac{3}{2}}}{(1+\alpha)(1-\alpha)^{\frac{1}{2}}}$

Q28. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

The value of K_C is 64 at 800 K for the reaction

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

The value of K_C for the following reaction is :

$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$

(A) 8 (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{64}$

Q29. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

For the reaction

$Fe_2N(s) + \frac{3}{2}H_2(g) \rightleftharpoons 2Fe(s) + NH_3(g)$

the correct relation between K_C and K_p is :

(A) $K_C = K_p(RT)^{1/2}$ (B) $K_C = K_p(RT)^{-1/2}$ (C) $K_C = K_p(RT)$ (D) $K_C = K_p(RT)^{3/2}$

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Q30. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

The variation of equilibrium constant with temperature is given below :

Temperature	Equilibrium Constant
$T_1 = 25^oC$	$K_1 = 10$
$T_2 = 100^oC$	$K_2 = 100$

The values of ΔH^o, ΔG^o at
T_1 and ΔG^o at T_2 (in kJ mol^{-1}) respectively, are close to :
[Use R = 8.314 J K^{-1} mol^{-1}]

(A) 28.4, −5.71 and −14.29

(B) 0.64, −7.14 and −5.71

(C) 28.4, −7.14 and −5.71

(D) 0.64, −5.71 and −14.29

Q31. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

Consider the following reaction:

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$; ΔH^o = +58 kJ

For each of the following cases (a, b), the direction in which the equilibrium shifts is :
(a) Temperature is decreased.
(b) Pressure is increased by adding N_2 at constant T.

(A) (a) towards reactant, (b) towards product

(B) (a) towards reactant, (b) no change

(C) (a) towards product, (b) towards reactant

(D) (a) towards product, (b) no change

Q32. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

If the equilibrium constant for
A ⇌ B + C is $K_{eq}^{(1)}$ and that of
B + C ⇌ P is $K_{eq}^{(2)}$, the equilibrium
constant for A ⇌ P is :

(A) $\frac{K_{eq}^{(1)}}{K_{eq}^{(2)}}$

(B) $K_{eq}^{(1)} + K_{eq}^{(2)}$

(C) $K_{eq}^{(2)} - K_{eq}^{(1)}$

(D) $K_{eq}^{(1)}\ K_{eq}^{(2)}$

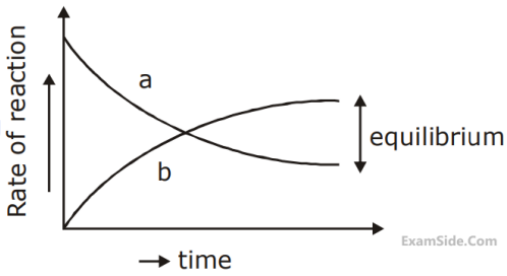
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Q33. JEE Main 2020 (Online) 4th September Morning Slot

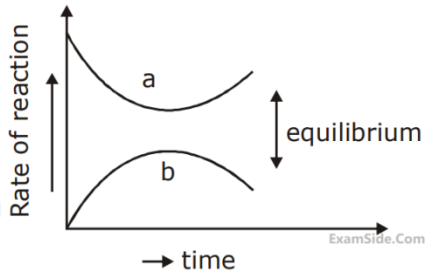
Open solution

For the equilibrium A ⇌ B , the variation of the rate of the forward (a) and reverse (b) reaction with time is given by :

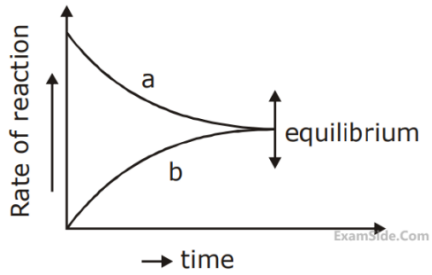
(A)



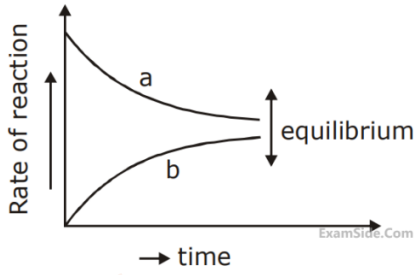
(B)



(C)



(D)

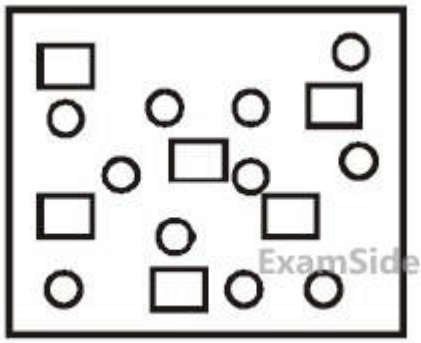


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Q34. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

In the figure shown below reactant A (represented by square) is in equilibrium with product B (represented by circle). The equilibrium constant is :



(A) 1 (B) 2 (C) 8 (D) 4

Q35. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

The INCORRECT match in the following is :

(A) $\Delta G^o = 0, K = 1$

(B) $\Delta G^o < 0, K < 1$

(C) $\Delta G^o > 0, K < 1$

(D) $\Delta G^o < 0, K > 1$

Q36. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

In which one of the following equilibria, $K_p \neq K_C$?

(A) $2NO(g) \leftrightharpoons N_2(g) + O_2(g)$

(B) $2C(s) + O_2(g) \leftrightharpoons 2CO(g)$

(C) $2HI(g) \leftrightharpoons H_2(g) + I_2(g)$

(D) $NO_2(g) + SO_2(g) \leftrightharpoons NO(g) + SO_3(g)$

Q37. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

For the reaction,

$2SO_2(g) + O_2(g) = 2SO_3(g)$, $\Delta H = -57.2$ kJ mol^{-1} and $K_C = 1.7 \times 10^{16}$

Which of the following statement is incorrect ?

(A) The equilibrium will shift in forward direction as the pressure increase.

(B) The addition of inert gas at constant volume will be not affect the equilibrium constant.

(C) The equilibrium constant is large suggestive of reaction going to completion and so no catalyst is required.

(D) The equilibrium constant decreases as the temperature increase.

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Q38. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

For the following reactions, equilibrium constants are given :

$S(s) + O_2(g) \leftrightharpoons SO_2(g)$; $K_1 = 10^{52}$

$2S(s) + 3O_2(g) \leftrightharpoons 2SO_3(g)$; $K_2 = 10^{129}$

The equilibrium constant for the reaction,

$2SO_2(g) + O_2(g) \leftrightharpoons 2SO_3(g)$ is :

(A) 10^{181} (B) 10^{25} (C) 10^{77} (D) 10^{154}

Q39. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

In a chemical reaction,

$$A + 2B \underset{}{\overset{K}{\rightleftharpoons}} 2C + D$$

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the initial concentration of B was 1.5 times of the concentration of A, but the equilibrium concentrations of A and B were found to be equal. The equilibrium constant (K) for the aforesaid chemical reaction is -

(A) 16 (B) 1 (C) 1/4 (D) 4

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Q40. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

Two solids dissociate as follows –

$$A(s) \rightleftharpoons B(g) + C(g) \; ; \; K_{p1} = x \text{ atm}^2$$

$$D(s) \rightleftharpoons C(g) + E(g) \; ; \; K_{p2} = y \text{ atm}^2$$

The total pressure when both the solids dissociated simultaneously is -

(A) (x + y) atm

(B) $\left(\sqrt{x+y}\right)$ atm

(C) $2\left(\sqrt{x+y}\right)$ atm

(D) $x^2 + y^2$ atm

Q41. JEE Main 2019 (Online) 11th January Morning Slot

Open solution

Consider the reaction

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

The equilibrium constant of the above reaction is K_p. If pure ammonia is left to dissociate, the partial pressure of ammonia at equilibrium is given by (Assume that $P_{NH_3} << P_{total}$ at equilibrium)

(A) $\frac{3^{\frac{3}{2}} K_P^{\frac{1}{2}} P^2}{4}$

(B) $\frac{K_P^{\frac{1}{2}} P^2}{4}$

(C) $\frac{3^{\frac{3}{2}} K_P^{\frac{1}{2}} P^2}{16}$

(D) $\frac{K_P^{\frac{1}{2}} P^2}{16}$

Q42. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

5.1 g NH_4SH is introduced in 3.0 L evacuated flask at 327°C. 30% of the solid NH_4SH decomposed to NH_3 and H_2S as gases . The Kp of the reaction at 327°C is (R = 0.082 L atm mol^{-1} K^{-1}, Molar mass of S = 32 g mol^{-1} molar mass of N = 14 g mol^{-1})

(A) 0.242×10^{-4} atm^2

(B) 1×10^{-4} atm^2

(C) 4.9×10^{-3} atm^2

(D) 0.242 atm^2

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Q43. JEE Main 2019 (Online) 10th January Morning Slot

Open solution

The values of K_p/K_c for the following reactions at 300 K are, respectively : (At 300 K, RT = 24.62 dm^3 atm mol^{-1})

$N_2(g) + O_2(g) \rightleftharpoons 2\,NO(g)$

$N_2O_4(g) \rightleftharpoons 2\,NO_2(g)$

$N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3(g)$

(A) 24.62 dm^3 atm mol^{-1}, 606.0 dm^6 atm^2 mol^{-2}, 1.65×10^{-3} dm^{-6} atm^{-2} mol^2

(B) 1,4.1 $\times 10^{-2}$ dm^{-3} atm^{-1} mol, 606 dm^6 atm^2 mol^{-2}

(C) 1,24.62 dm^3 atm mol^{-1} , 1.65×10^{-3} dm^{-6} atm $^{-2}$ mol^2

(D) 1, 24.62 dm^3 atm mol^{-1}, 606.0 dm^6 atm^2 mol^{-2}

Q44. JEE Main 2019 (Online) 9th January Evening Slot

Open solution

Consider the following reversible chemical reactions :

$$A_2(g) + B_2(g) \underset{}{\overset{K_1}{\rightleftharpoons}} 2AB(g) \ldots\ldots (1)$$

$$6AB(g) \underset{}{\overset{K_2}{\rightleftharpoons}} 3A_2(g) + 3B_2(g) \ldots\ldots (2)$$

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The relation between K_1 and K_2 is :

(A) $K_1K_2 = \frac{1}{3}$ (B) $K_2 = {K_1}^3$ (C) $K_2 = {K_1}^{-3}$ (D) $K_1K_2 = 3$

Q45. JEE Main 2018 (Online) 16th April Morning Slot

Open solution

The gas phase reaction $2NO_2(g) \rightarrow N_2O_4(g)$ is an exothermic reaction. The decomposition of N_2O_4, in equilibrium mixture of $NO_2(g)$ and $N_2O_4(g)$, can be increased by :

(A) lowering the temperature.

(B) increasing the pressure.

(C) addition of an inert gas at constant volume.

(D) addition of an inert gas at constant pressure.

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Q46. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

At a certain temperature in a $5\ L$ vessel, 2 moles of carbon monoxide and 3 moles of chlorine were allowed to reach equilibrium according to the reaction,

$$CO + Cl_2 \rightleftharpoons COCl_2$$

At equilibrium, if one mole of CO is present then equilibrium constant (K_c) for the reaction is :

(A) 2 (B) 2.5 (C) 3 (D) 4

Q47. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

In which of the following reactions, an increase in the volume of the container will favour the formation of products?

(A) $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$

(B) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(C) $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(l)$

(D) $3O_2\ (g) \rightleftharpoons 2O_3(g)$

Q48. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal :

$$Fe_2O_3(s) + 3\ CO(g) \rightleftharpoons 2\ Fe(l) + 3\ CO_2(g)$$

Using the Le Chatelier's principle, predict which one of the following will not disturb the equilibrium ?

(A) Removal of CO (B) Removal of CO_2

(C) Addition of CO_2 (D) Addition of Fe_2O_3

Q49. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

A solid XY kept in an evacuated sealed container undergoes decomposition to form a mixture of gases X and Y at temperature T. The equilibrium pressure is 10 bar in this vessel. K_p for this reaction is :

(A) 5 (B) 10 (C) 25 (D) 100

Q50. JEE Main 2016 (Offline)

Open solution

The equilibrium constant at 298 K for a reaction A + B $\rightleftharpoons$ C + D is 100. If the initial concentration of all the four species were 1M each, then equilibrium concentration of D (in mol L^{-1}) will be:

(A) 0.818 (B) 1.818 (C) 1.182 (D) 0.182

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Q51. JEE Main 2015 (Online) 11th April Morning Slot

The increase of pressure on ice $\rightleftharpoons$ water system at constant temperature will lead to :

(A) no effect on the equilibrium

(B) a decrease in the entropy of the system

(C) a shift of the equilibrium in the forward direction

(D) an increase in the Gibbs energy of the system

Open solution

Q52. JEE Main 2015 (Online) 10th April Morning Slot

Gaseous N_2O_4 dissociates into gaseous NO_2 according to the reaction

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

At 300 K and 1 atm pressure, the degree of dissociation of N_2O_4 is 0.2. If one mole of N_2O_4 gas is contained in a vessel, then the density of the equilibrium mixture is :

(A) 1.56 g/L (B) 3.11 g/L (C) 4.56 g/L (D) 6.22 g/L

Open solution

Q53. JEE Main 2015 (Offline)

The standard Gibbs energy change at 300 K for the reaction $2A \rightleftharpoons B + C$ is 2494.2 J. At a given time, the composition of the reaction mixture is [A] = 1/2, [B] = 2 and [C] = 1/2. The reaction proceeds in the: [R = 8.314 J/K/mol, e = 2.718]

(A) reverse direction because $Q > K_c$

(B) forward direction because $Q < K_c$

(C) reverse direction because $Q < K_c$

(D) forward direction because $Q > K_c$

Open solution

Q54. JEE Main 2014 (Online) 19th April Morning Slot

For the decomposition of the compound represented as

$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

the $K_p = 2.9 \times 10^{-5}$ atm^3.

If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be :

(A) 1.94×10^{-2} atm

(B) 5.82×10^{-2} atm

(C) 7.66×10^{-2} atm

(D) 38.8×10^{-2} atm

Open solution

Q55. JEE Main 2014 (Online) 12th April Morning Slot

What happens when an inert gas is added to an equilibrium keeping volume unchanged ?

(A) More product will form

(B) Less product will form

(C) More reactant will form

(D) Equilibrium will remain unchanged

Open solution

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Q56. JEE Main 2014 (Online) 11th April Morning Slot

Consider the following equilibrium

$AgCl\downarrow + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$

White precipitate of AgCl appears on adding which of the following ?

(A) NH_3 (B) aqueous NaCl (C) aqueous HNO_3 (D) aqueous NH_4Cl

Open solution

Q57. JEE Main 2014 (Online) 9th April Morning Slot

At a certain temperature, only 50% HI is dissociated into H_2 and I_2 at equilibrium. The equilibrium constant is :

(A) 1.0 (B) 3.0 (C) 0.5 (D) 0.25

Open solution

Q58. JEE Main 2014 (Offline)

For the reaction SO_2 (g) + $\frac{1}{2}O_2(g) \rightleftharpoons SO_3$(g).

if $K_P = K_C(RT)^X$ where the symbols have usual meaning then the value of x is: (assuming ideality)

(A) -1 (B) -1/2 (C) 1/2 (D) 1

Open solution

Q59. JEE Main 2013 (Online) 25th April Morning Slot

The ratio $\frac{K_p}{K_c}$ for the reaction $CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$ is:

(A) $\frac{1}{\sqrt{RT}}$ (B) $(RT)^{1/2}$ (C) RT (D) 1

Open solution

Q60. JEE Main 2013 (Online) 23th April Morning Slot

(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, K_1
(2) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, K_2
(3) $H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g)$, K_3
The equation for the equilibrium constant of the reaction $2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$, (K_4) in terms of K_1, K_2 and K_3 is :

(A) $\frac{K_1 \cdot K_2}{K_3}$ (B) $\frac{K_1 \cdot K_2^2}{K_2}$ (C) $K_1K_2K_3$ (D) $\frac{K_2 \cdot K_3^3}{K_1}$

Open solution

Q61. JEE Main 2013 (Online) 9th April Morning Slot

In reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of B was 1.5 times of [A], but at equilibrium the concentrations of A and B became equal. The equilibrium constant for the reaction is :

(A) 8 (B) 4 (C) 12 (D) 6

Open solution

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Q62. AIEEE 2012 (Online) 26th May

One mole of $O_{2(g)}$ and two moles of $SO_{2(g)}$ were heated in a closed vessel of one-litre capacity at $1098\ K$. At equilibrium 1.6 moles of $SO_{3(g)}$ were found. The equilibrium constant K_c of the reaction would be

Open solution

(A) 30 (B) 40 (C) 80 (D) 60

Q63. AIEEE 2012 (Online) 19th May

The value of K_p for the equilibrium reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 2. The percentage dissociation of $N_2O_4(g)$ at a pressure of $0.5\ atm$ is

Open solution

(A) 25 (B) 88 (C) 50 (D) 71

Q64. AIEEE 2012 (Online) 12th May

$8\ mol$ of $AB_3(g)$ are introduced into a $1.0\ dm^3$ vessel. If it dissociates as $2AB_3(g) \rightleftharpoons A_2(g) + 3B_2(g)$. At equilibrium, $2\ mol$ of A_2 are found to be present. The equilibrium constant of this reaction is

Open solution

(A) 2 (B) 3 (C) 27 (D) 36

Q65. AIEEE 2012 (Online) 7th May

K_1, K_2 and K_3 are the equilibrium constants of the following reactions (I), (II) and (III) respectively:
(I) $N_2 + 2O_2 \rightleftharpoons 2NO_2$
(II) $2NO_2 \rightleftharpoons N_2 + 2O_2$
(III) $NO_2 \rightleftharpoons \frac{1}{2}N_2 + O_2$
The correct relation from the following is

Open solution

(A) $K_1 = \frac{1}{K_2} = \frac{1}{K_3}$ (B) $K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$
(C) $K_1 = \sqrt{K_2} = K_3$ (D) $K_1 = \frac{1}{K_2} = K_3$

Q66. AIEEE 2012

The equilibrium constant (K_C) for the reaction $N_2(g) + O_2(g) \rightarrow 2NO(g)$ at temperature T is 4×10^{-4}. The value of K_C for the reaction, $NO(g) \rightarrow 1/2N_2(g) + 1/2O_2(g)$ at the same temperature is :

Open solution

(A) 0.02 (B) 2.5×10^2 (C) 4×10^{-4} (D) 50.0

Q67. AIEEE 2011

A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K is :

Open solution

(A) 3 atm (B) 0.3 atm (C) 0.18 atm (D) 1.8 atm

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Q68. AIEEE 2008

The equilibrium constants K_{P1} and K_{P2} for the reactions X $\leftrightharpoons$ 2Y and Z $\leftrightharpoons$ P + Q, respectively are in the ratio of 1 : 9. If the degree of dissociation of X and Z be equal then the ratio of total pressure at these equilibria is :

Open solution

(A) 1 : 36 (B) 1 : 1 (C) 1 : 3 (D) 1 : 9

Q69. AIEEE 2008

For the following three reactions a, b and c, equilibrium constants are given:

a. $CO\ (g) + H_2O\ (g) \leftrightharpoons CO_2(g) + H_2\ (g)$; K_1

b. $CH_4\ (g) + H_2O\ (g) \leftrightharpoons CO(g) + 3H_2\ (g)$; K_2

c. $CH_4\ (g) + 2H_2O\ (g) \leftrightharpoons CO_2(g) + 4H_2\ (g)$; K_3

Which of the following relations is correct?

Open solution

(A) $K_1\sqrt{K_2} = K_3$ (B) $K_2K_3 = K_1$ (C) $K_3 = K_1K_2$ (D) $K_3.K_2^3 = K_1^2$

Q70. AIEEE 2006

Phosphorus pentachloride dissociates as follows, in a closed reaction vessel

$PCl_5\ (g) \leftrightharpoons PCl_3\ (g) + Cl_2\ (g)$

If total pressure at equilibrium of the reaction mixture is P and degree of dissociation of PCl_5 is x, the partial pressure of PCl_3 will be

Open solution

(A) $\left(\frac{x}{x+1}\right)P$ (B) $\left(\frac{2x}{1-x}\right)P$

(C) $\left(\frac{x}{x-1}\right)P$ (D) $\left(\frac{x}{1-x}\right)P$

Q71. AIEEE 2006

The equilibrium constant for the reaction

$SO_3\ (g) \leftrightharpoons SO_2\ (g) + \frac{1}{2}\ O_2\ (g)$

is $K_c = 4.9 \times 10^{-2}$. The value of K_c for the reaction

$2SO_2\ (g) + O_2\ (g) \leftrightharpoons 2SO_3\ (g)$ will be :

Open solution

(A) 416 (B) 9.8×10^{-2} (C) 4.9×10^{-2} (D) 2.40×10^{-3}

Q72. AIEEE 2005

The exothermic formation of ClF_3 is represented by the equation:

$Cl_2\ (g) + 3F_2\ (g) \leftrightharpoons 2ClF_3\ (g)$; ΔH = -329 kJ

Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2, F_2 and ClF_3?

Open solution

(A) Increasing the temperature (B) Removing Cl_2

(C) Increasing the volume of the container (D) Adding F_2

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Q73. AIEEE 2005

For the reaction $2NO_2$ (g) $\rightleftharpoons$ 2NO (g) + O_2 (g), ($K_c = 1.8 \times 10^{-6}$ at 184°C) (R = 0.0831 kJ/(mol. K))
When K_p and K_c are compared at 184°C , it is found that :

(A) K_p is greater than K_c

(B) K_p is less than K_c

(C) $K_p = K_c$

(D) Whether K_p is greater than, less than or equal to K_c depends upon the total gas pressure

Open solution

Q74. AIEEE 2005

An amount of solid NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. Ammonium hydrogen sulphide decomposes to yield NH_3 and H_2S gases in the flask. When the decomposition reaction reaches equilibrium, the total pressure in the flask rises to 0.84 atm. The equilibrium constant for NH_4HS decomposition at this temperature is :

(A) 0.30 (B) 0.11 (C) 0.17 (D) 0.18

Open solution

Q75. AIEEE 2004

What is the equilibrium expression for the reaction
P_4 (s) + $5O_2 \rightleftharpoons P_4O_{10}$ (s)?

(A) $K_c = [P_4O_{10}] / 5[P_4] [O_2]$ (B) $K_c = 1/[O_2]^5$

(C) $K_c = [P_4O_{10}] / [P_4] [O_2]^5$ (D) $K_c = [O_2]^5$

Open solution

Q76. AIEEE 2004

For the reaction, CO(g) + Cl_2(g) $\rightleftharpoons$ $COCl_2$(g) the $\frac{K_p}{K_c}$ is equal to :

(A) $\sqrt{RT}$ (B) RT (C) 1/RT (D) 1.0

Open solution

Q77. AIEEE 2004

The equilibrium constant for the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ at temperature T is 4×10^{-4}. The value of K_c for the reaction $NO(g) \rightleftharpoons \frac{1}{2}N_2$ (g) + $\frac{1}{2}O_2$ (g) at the same temperature is :

(A) 2.5×10^2 (B) 4×10^{-4} (C) 50 (D) 0.02

Open solution

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Q78. AIEEE 2003

For the reaction equilibrium
N_2O_4 (g) $\rightleftharpoons$ $2NO_2$ (g)
the concentrations of N_2O_4 and NO_2 at equilibrium are 4.8×10^{-2} and 1.2×10^{-2} mol L^{-1} respectively. The value of K_c for the reaction is

(A) 3×10^{-1} mol L^{-1}
(B) 3×10^{-3} mol L^{-1}
(C) 3×10^{3} mol L^{-1}
(D) 3.3×10^{2} mol L^{-1}

Open solution

Q79. AIEEE 2003

Consider the reaction equilibrium
2 SO_2 (g) + O_2 (g) $\rightleftharpoons$ 2 SO_3 (g); ΔH^o = -198 kJ
One the basis of Le Chatelier's principle, the condition favourable for the forward reaction is :

(A) increasing temperature as well as pressure
(B) lowering the temperature and increasing the pressure
(C) any value of temperature and pressure
(D) lowering temperature as well as pressure

Open solution

Q80. AIEEE 2002

Change in volume of the system does not alter which of the following equilibria?

(A) $N_2(g) + O_2(g) \rightleftharpoons 2NO$ (g)
(B) $PCl_5(g) \rightleftharpoons PCl_3$ (g) + Cl_2 (g)
(C) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3$ (g)
(D) SO_2Cl_2 (g) $\rightleftharpoons$ SO_2 (g) + Cl_2 (g)

Open solution

Q81. AIEEE 2002

For the reaction CO (g) + (1/2) O_2 (g) $\rightleftharpoons$ CO_2 (g), K_p/K_c is :

(A) RT
(B) $(RT)^{-1}$
(C) $(RT)^{-1/2}$
(D) $(RT)^{1/2}$

Open solution

Numerical 49 questions

Q82. JEE Main 2026 (Online) 8th April Evening Shift

Solid carbon, CaO and $CaCO_3$ are mixed and allowed to attain equilibrium at T K .
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g) \quad Kp_1 = 0.08\ atm$
$C(s) + CO_2(g) \rightleftharpoons 2CO(g) \quad Kp_2 = 2\ atm$
The partial pressure of CO is __ $\times 10^{-1}$ atm

Open solution

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Q83. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

In a closed flask at 600 K , one mole of $X_2Y_4(g)$ attains equilibrium as given below :

$X_2Y_4(g) \rightleftharpoons 2XY_2(g)$

At equilibrium, $75\% X_2Y_4(g)$ was dissociated and the total pressure is 1 atm . The magnitude of $\Delta_r G^{\ominus}$ (in $kJmol^{-1}$) at this temperature is ____ . (Nearest Integer)

(Given : $R = 8.3\ J\ mol^{-1}\ K^{-1}; \ln 10 = 2.3, \log 2 = 0.3, \log 3 = 0.48, \log 5 = 0.69, \log 7 = 0.84$)

Q84. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

The values of pressure equilibrium constant recorded at different temperatures for the following equilibrium reaction have been given below $A(g) \rightleftharpoons B(g) + C(g)$

$\frac{1}{T}\left(K^{-1}\right)$	$\log_{10} K_p$
0.05	3.5
0.06	2.5
0.07	1.5

The magnitude of $\frac{\Delta H^{\circ}}{R}$ calculated from the above data is ____ . (Nearest integer)

Q85. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

For the following reaction at $50^{\circ}C$ and at 2 atm pressure,

$2\,N_2O_5(g) \rightleftharpoons 2\,N_2O_4(g) + O_2(g)$

N_2O_5 is 50% dissociated.

The magnitude of standard free energy change at this temperature is x.

$x =$ ____ $Jmol^{-1}$ [Nearest integer].

Given : $R = 8.314\ J\ mol^{-1}\ K^{-1}, \log 2 = 0.30, \log 3 = 0.48, \ln 10 = 2.303, {}^{\circ}C + 273 = K$

Q86. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

$X_2(g) + Y_2(g) \rightleftharpoons 2Z(g)$

$X_2(g)$ and $Y_2(g)$ are added to a 1 L flask and it is found that the system attains the above equilibrium at $T(K)$ with the number of moles of $X_2(g), Y_2(g)$ and $Z(g)$ being 3,3 and 9 mol respectively (equilibrium moles). Under this condition of equilibrium, 10 mol of $Z(g)$ is added to the flask and the temperature is maintained at $T(K)$. Then the number of moles of $Z(g)$ in the flask when the new equilibrium is established is ____ . (Nearest integer)

Q87. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

For the following gas phase equilibrium reaction at constant temperature,

$NH_3(g) \rightleftharpoons 1/2\,N_2(g) + 3/2H_2(g)$

if the total pressure is $\sqrt{3}\,atm$ and the pressure equilibrium constant (K_p) is 9 atm , then the degree of dissociation is given as $\left(x \times 10^{-2}\right)^{-1/2}$. The value of x is ____ . (nearest integer)

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Q88. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Dissociation of a gas A_2 takes place according to the following chemical reaction. At equilibrium, the total pressure is 1 bar at 300 K .

$A_2(g) \rightleftharpoons 2\,A(g)$

The standard Gibbs energy of formation of the involved substances has been provided below:

Substance	$\Delta G_f^\circ / \text{kJ mol}^{-1}$
A_2	-100.00
A	-50.832

The degree of dissociation of $A_2(g)$ is given by $\left(x \times 10^{-2}\right)^{1/2}$ where $x =$ ____ . (Nearest integer).

[**Given :** $R = 8\ \text{J mol}^{-1}\ \text{K}^{-1}, \log 2 = 0.3010, \log 3 = 0.48$]

Assume degree of dissociation is not negligible.

Q89. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

The equilibrium constant for decomposition of $H_2O(g)$

$H_2O(g) \rightleftharpoons H_2(g) + \frac{1}{2}O_2(g) \quad (\Delta G^\circ = 92.34\ \text{kJ mol}^{-1})$

is 8.0×10^{-3} at 2300 K and total pressure at equilibrium is 1 bar. Under this condition, the degree of dissociation (α) of water is _____ $\times 10^{-2}$ (nearest integer value).

[Assume α is negligible with respect to 1]

Q90. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

Consider the following equilibrium,

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

0.1 mol of CO along with a catalyst is present in a 2dm^3 flask maintained at 500 K . Hydrogen is introduced into the flask until the pressure is 5 bar and 0.04 mol of CH_3OH is formed. The K_p^θ is _____ $\times 10^{-3}$ (nearest integer).

Given : $R = 0.08\ \text{dm}^3$ bar $\text{K}^{-1}\ \text{mol}^{-1}$

Assume only methanol is formed as the product and the system follows ideal gas behaviour.

Q91. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

$37.8\ \text{g}\ N_2O_5$ was taken in a 1 L reaction vessel and allowed to undergo the following reaction at 500 K

$2\,N_2O_{5(g)} \rightleftharpoons 2\,N_2O_{4(g)} + O_{2(g)}$

The total pressure at equilibrium was found to be 18.65 bar.

Then, Kp = _____ $\times 10^{-2}$ [nearest integer]

Assume N_2O_5 to behave ideally under these conditions.

Given: $R = 0.082$ bar $\text{Lmol}^{-1}\ \text{K}^{-1}$

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Q92. JEE Main 2024 (Online) 29th January Evening Shift

The following concentrations were observed at $500\ \mathrm{K}$ for the formation of $\mathrm{NH_3}$ from $\mathrm{N_2}$ and $\mathrm{H_2}$. At equilibrium ; $[\mathrm{N_2}] = 2 \times 10^{-2}\mathrm{M}$, $[\mathrm{H_2}] = 3 \times 10^{-2}\mathrm{M}$ and $[\mathrm{NH_3}] = 1.5 \times 10^{-2}\mathrm{M}$. Equilibrium constant for the reaction is _____.

Open solution

Q93. JEE Main 2024 (Online) 29th January Morning Shift

For the reaction $\mathrm{N_2O_{4(g)}} \rightleftarrows 2\mathrm{NO_{2(g)}}, \mathrm{K_p} = 0.492\ \mathrm{atm}$ at $300\ \mathrm{K}$. $\mathrm{K_c}$ for the reaction at same temperature is _____ $\times 10^{-2}$.

(Given : $\mathrm{R} = 0.082\ \mathrm{L\ atm\ mol^{-1}\ K^{-1}}$)

Open solution

Q94. JEE Main 2023 (Online) 11th April Evening Shift

4.5 moles each of hydrogen and iodine is heated in a sealed ten litre vessel. At equilibrium, 3 moles of HI were found. The equilibrium constant for $\mathrm{H_2(g)} + \mathrm{I_2(g)} \rightleftharpoons 2\mathrm{HI(g)}$ is _____.

Open solution

Q95. JEE Main 2023 (Online) 11th April Morning Shift

A mixture of 1 mole of $\mathrm{H_2O}$ and 1 mole of CO is taken in a 10 litre container and heated to $725\ \mathrm{K}$. At equilibrium 40% of water by mass reacts with carbon monoxide according to the equation :

$$\mathrm{CO(g)} + \mathrm{H_2O(g)} \rightleftharpoons \mathrm{CO_2(g)} + \mathrm{H_2(g)}.$$

The equilibrium constant $\mathrm{K_c} \times 10^2$ for the reaction is _____. (Nearest integer)

Open solution

Q96. JEE Main 2023 (Online) 10th April Evening Shift

$$\mathrm{A}(g) \rightleftharpoons 2\,\mathrm{B}(g) + \mathrm{C}(g)$$

For the given reaction, if the initial pressure is $450\ \mathrm{mm\ Hg}$ and the pressure at time t is $720\ \mathrm{mm\ Hg}$ at a constant temperature T and constant volume V. The fraction of $\mathrm{A(g)}$ decomposed under these conditions is $x \times 10^{-1}$. The value of x is _____ (nearest integer)

Open solution

Q97. JEE Main 2023 (Online) 10th April Morning Shift

The number of correct statement/s involving equilibria in physical processes from the following is _____

(A) Equilibrium is possible only in a closed system at a given temperature.

(B) Both the opposing processes occur at the same rate.

(C) When equilibrium is attained at a given temperature, the value of all its parameters became equal.

(D) For dissolution of solids in liquids, the solubility is constant at a given temperature.

Open solution

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Q98. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

The equilibrium composition for the reaction $PCl_3 + Cl_2 \rightleftharpoons PCl_5$ at $298\ K$ is given below:

$[PCl_3]_{eq} = 0.2\ mol\ L^{-1}, [Cl_2]_{eq} = 0.1\ mol\ L^{-1}, [PCl_5]_{eq} = 0.40\ mol\ L^{-1}$

If $0.2\ mol$ of Cl_2 is added at the same temperature, the equilibrium concentration of PCl_5 is _____ $\times 10^{-2}\ mol\ L^{-1}$

Given : K_c for the reaction at $298\ K$ is 20

Q99. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

(i) $X(g) \rightleftharpoons Y(g) + Z(g) \quad K_{p1} = 3$

(ii) $A(g) \rightleftharpoons 2\,B(g) \quad K_{p2} = 1$

If the degree of dissociation and initial concentration of both the reactants $X(g)$ and $A(g)$ are equal, then the ratio of the total pressure at equilibrium $\left(\frac{p_1}{p_2}\right)$ is equal to $x : 1$. The value of x is _____ (Nearest integer)

Q100. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

For reaction : $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$

$K_p = 2 \times 10^{12}$ at $27^\circ C$ and $1\ atm$ pressure. The K_c for the same reaction is _____ $\times 10^{13}$. (Nearest integer)

(Given $R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}$)

Q101. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

Consider the following equation:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g), \Delta H = -190\ kJ$

The number of factors which will increase the yield of SO_3 at equilibrium from the following is _____.

A. Increasing temperature

B. Increasing pressure

C. Adding more SO_2

D. Adding more O_2

E. Addition of catalyst

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Q102. JEE Main 2023 (Online) 29th January Evening Shift

At 298 K

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g),\ K_1 = 4 \times 10^5$

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g),\ K_2 = 1.6 \times 10^{12}$

$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g),\ K_3 = 1.0 \times 10^{-13}$

Based on above equilibria, then equilibrium constant of the reaction,
$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$ is _____ $\times 10^{-33}$ (Nearest integer).

Open solution

Q103. JEE Main 2023 (Online) 29th January Morning Shift

Water decomposes at 2300 K

$H_2O(g) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$

The percent of water decomposing at 2300 K and 1 bar is _____ (Nearest integer).

Equilibrium constant for the reaction is 2×10^{-3} at 2300 K.

Open solution

Q104. JEE Main 2023 (Online) 29th January Morning Shift

Consider the following reaction approaching equilibrium at 27°C and 1 atm pressure

$$A + B \underset{k_r=10^2}{\overset{k_f=10^3}{\rightleftharpoons}} C + D$$

The standard Gibb's energy change $(\Delta_r G^\theta)$ at 27°C is (–) _____ kJ mol^{-1} (Nearest integer).

(Given : $R = 8.3\ J\ K^{-1}\ mol^{-1}$ and $\ln 10 = 2.3$)

Open solution

Q105. JEE Main 2022 (Online) 28th July Evening Shift

At $600\ K, 2\ mol$ of NO are mixed with $1\ mol$ of O_2.

$2NO_{(g)} + O_2(g) \rightleftarrows 2NO_2(g)$

The reaction occurring as above comes to equilibrium under a total pressure of 1 atm. Analysis of the system shows that $0.6\ mol$ of oxygen are present at equilibrium. The equilibrium constant for the reaction is _____. (Nearest integer)

Open solution

Q106. JEE Main 2022 (Online) 26th July Morning Shift

At $298\ K$, the equilibrium constant is 2×10^{15} for the reaction :

$Cu(s) + 2Ag^+(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$

The equilibrium constant for the reaction

$\frac{1}{2}Cu^{2+}(aq) + Ag(s) \rightleftharpoons \frac{1}{2}Cu(s) + Ag^+(aq)$

is $x \times 10^{-8}$. The value of x is _____. (Nearest Integer)

Open solution

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Q107. JEE Main 2022 (Online) 29th June Evening Shift

A box contains 0.90 g of liquid water in equilibrium with water vapour at 27°C. The equilibrium vapour pressure of water at 27°C is 32.0 Torr. When the volume of the box is increased, some of the liquid water evaporates to maintain the equilibrium pressure. If all the liquid water evaporates, then the volume of the box must be _____ litre. [nearest integer]

(Given : R = 0.082 L atm K^{-1} mol^{-1})

(Ignore the volume of the liquid water and assume water vapours behave as an ideal gas.)

Open solution

Q108. JEE Main 2022 (Online) 27th June Morning Shift

$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

In an experiment, 2.0 moles of NOCl was placed in a one-litre flask and the concentration of NO after equilibrium established, was found to be 0.4 mol/L. The equilibrium constant at 30°C is _____ $\times 10^{-4}$.

Open solution

Q109. JEE Main 2022 (Online) 26th June Evening Shift

40% of HI undergoes decomposition to H_2 and I_2 at 300 K. $\Delta G^{\ominus}$ for this decomposition reaction at one atmosphere pressure is _____ J mol^{-1}. [nearest integer]

(Use R = 8.31 J K^{-1} mol^{-1} ; log 2 = 0.3010, ln 10 = 2.3, log 3 = 0.477)

Open solution

Q110. JEE Main 2022 (Online) 25th June Morning Shift

The standard free energy change (ΔG°) for 50% dissociation of N_2O_4 into NO_2 at 27°C and 1 atm pressure is $-$ x J mol^{-1}. The value of x is _____. (Nearest Integer)

[Given : R = 8.31 J K^{-1} mol^{-1}, log 1.33 = 0.1239, ln 10 = 2.3]

Open solution

Q111. JEE Main 2022 (Online) 24th June Evening Shift

PCl_5 dissociates as

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

5 moles of PCl_5 are placed in a 200 litre vessel which contains 2 moles of N_2 and is maintained at 600 K. The equilibrium pressure is 2.46 atm. The equilibrium constant K_p for the dissociation of PCl_5 is _____ $\times 10^{-3}$. (nearest integer)

(Given : R = 0.082 L atm K^{-1} mol^{-1}; Assume ideal gas behaviour)

Open solution

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Q112. JEE Main 2022 (Online) 24th June Morning Shift

$2O_3(g) \rightleftharpoons 3O_2(g)$

At 300 K, ozone is fifty percent dissociated. The standard free energy change at this temperature and 1 atm pressure is (–) _____ J mol^{-1}. (Nearest integer)

[Given : ln 1.35 = 0.3 and R = 8.3 J K^{-1} mol^{-1}]

Open solution

Q113. JEE Main 2021 (Online) 27th August Evening Shift

When 5.1 g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C, 20% of the solid decomposes into gaseous ammonia and hydrogen sulphide. The K_p for the reaction at 27°C is x $\times$ 10^{-2}. The value of x is _____. (Integer answer) [Given R = 0.082 L atm K^{-1} mol^{-}]

Open solution

Q114. JEE Main 2021 (Online) 27th August Morning Shift

The number of moles of NH_3, that must be added to 2L of 0.80 M $AgNO_3$ in order to reduce the concentration of Ag^+ ions to 5.0 $\times$ 10^{-8} M ($K_{formation}$ for $[Ag(NH_3)_2]^+$ = 1.0 $\times$ 10^8) is _____. (Nearest integer)

[Assume no volume change on adding NH_3]

Open solution

Q115. JEE Main 2021 (Online) 26th August Evening Shift

The equilibrium constant K_c at 298 K for the reaction A + B $\rightleftharpoons$ C + D is 100. Starting with an equimolar solution with concentrations of A, B, C and D all equal to 1M, the equilibrium concentration of D is _____ $\times$ 10^{-2} M. (Nearest integer)

Open solution

Q116. JEE Main 2021 (Online) 26th August Evening Shift

The reaction rate for the reaction

$[PtCl_4]^{2-} + H_2O \rightleftharpoons [Pt(H_2O)Cl_3]^- + Cl^-$

was measured as a function of concentrations of different species. It was observed that

$$\frac{-d\left[[PtCl_4]^{2-}\right]}{dt} = 4.8 \times 10^{-5}\left[[PtCl_4]^{2-}\right] - 2.4 \times 10^{-3}\left[[Pt(H_2O)Cl_3]^-\right][Cl^-].$$

where square brackets are used to denote molar concentrations. The equilibrium constant K_c = _____ . (Nearest integer)

Open solution

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Q117. JEE Main 2021 (Online) 27th July Evening Shift

The equilibrium constant for the reaction

$A(s) \rightleftharpoons M(s) + \frac{1}{2}O_2(g)$

is $K_p = 4$. At equilibrium, the partial pressure of O_2 is _____ atm. (Round off to the nearest integer)

Open solution

Q118. JEE Main 2021 (Online) 27th July Morning Shift

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$K_c = 1.844$

3.0 moles of PCl_5 is introduced in a 1 L closed reaction vessel at 380 K. The number of moles of PCl_5 at equilibrium is _____ $\times 10^{-3}$. (Round off to the Nearest Integer)

Open solution

Q119. JEE Main 2021 (Online) 25th July Morning Shift

For the reaction

$A + B \rightleftharpoons 2C$

the value of equilibrium constant is 100 at 298 K. If the initial concentration of all the three species is 1 M each, then the equilibrium concentration of C is $x \times 10^{-1}$ M. The value of x is _____. (Nearest integer)

Open solution

Q120. JEE Main 2021 (Online) 22th July Evening Shift

Value of K_P for the equilibrium reaction

$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ at 288 K is 47.9. The K_C for this reaction at same temperature is _____. (Nearest integer)

($R = 0.083$ L bar K^{-1} mol^{-1})

Open solution

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Q121. JEE Main 2021 (Online) 20th July Morning Shift

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

In an equilibrium mixture, the partial pressures are

P_{SO_3} = 43 kPa; P_{O_2} = 530 Pa and P_{SO_2} = 45 kPa. The equilibrium constant K_P = _____ $\times 10^{-2}$.
(Nearest integer)

Open solution

Q122. JEE Main 2021 (Online) 18th March Evening Shift

The gas phase reaction $2A(g) \rightleftharpoons A_2(g)$ at 400 K has ΔG^o = + 25.2 kJ mol^{-1}.

The equilibrium constant K_C for this reaction is _____ $\times 10^{-2}$. (Round off to the Nearest Integer).

[Use : R = 8.3 J mol^{-1} K^{-1}, ln 10 = 2.3, $\log_{10}$ 2 = 0.30, 1 atm = 1 bar]

[antilog (-0.3) = 0.501]

Open solution

Q123. JEE Main 2021 (Online) 17th March Evening Shift

Consider the reaction

$N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$

The temperature at which K_C = 20.4 and K_P = 600.1, is _____ K. (Round off to the Nearest Integer). [Assume all gases are ideal and R = 0.0831 L bar K^{-1} mol^{-1}]

Open solution

Q124. JEE Main 2021 (Online) 17th March Morning Shift

0.01 moles of a weak acid HA ($K_a = 2.0 \times 10^{-6}$) is dissolved in 1.0 L of 0.1 M HCl solution. The degree of dissociation of HA is _____ $\times 10^{-5}$ (Round off to the Nearest Integer).

[Neglect volume change on adding HA. Assume degree of dissociation <<1]

Open solution

Q125. JEE Main 2021 (Online) 16th March Morning Shift

For the reaction $A(g) \rightleftharpoons B(g)$ at 495 K, $\Delta_f G^\circ = -9.478$ kJ mol^{-1}.
If we start the reaction in a closed container at 495 K with 22 millimoles of A, the amount of B in the equilibrium mixture is _____ millimoles.
(Round off to the Nearest Integer). [R = 8.314 J mol^{-1} K^{-1}; ln 10 = 2.303]

Open solution

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Q126. JEE Main 2021 (Online) 26th February Morning Shift

A homogeneous ideal gaseous reaction $AB_{2(g)} \rightleftharpoons A_{(g)} + 2B_{(g)}$ is carried out in a 25 litre flask at 27°C. The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_p is $x \times 10^{-2}$. The value of x is _____. (Integer answer)

[R = 0.08206 dm^3atm $K^{-1}mol^{-1}$]

Open solution

Q127. JEE Main 2021 (Online) 24th February Morning Shift

The stepwise formation of $[Cu(NH_3)_4]^{2+}$ is given below:

$$Cu^{2+} + NH_3 \xrightleftharpoons{K_1} [Cu(NH_3)]^{2+}$$

$$[Cu(NH_3)]^{2+} + NH_3 \xrightleftharpoons{K_2} [Cu(NH_3)_2]^{2+}$$

$$[Cu(NH_3)_2]^{2+} + NH_3 \xrightleftharpoons{K_3} [Cu(NH_3)_3]^{2+}$$

$$[Cu(NH_3)_3]^{2+} + NH_3 \xrightleftharpoons{K_4} [Cu(NH_3)_4]^{2+}$$

The value of stability constants K_1, K_2, K_3 and K_4 are 10^4, 1.58 x 10^3, 5 x 10^2 and 10^2 respectively.
The overall equilibrium constants for dissociation of $[Cu(NH_3)_4]^{2+}$ is $x \times 10^{-12}$.
The value of x is _____. (Rounded off to the nearest integer)

Open solution

Q128. JEE Main 2021 (Online) 24th February Morning Shift

At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value of K_p for the reaction $Cl_{2\,(g)} \rightleftharpoons 2Cl_{(g)}$ under the above conditions is $x \times 10^{-1}$.
The value of x is _____. (Rounded off to the nearest integer)

Open solution

Q129. JEE Main 2021 (Online) 24th February Morning Shift

For the reaction $A_{(g)} \rightarrow B_{(g)}$ the value of the equilibrium constant at 300 K and 1 atm is equal to 100.0. The value of $\Delta_r G$ for the reaction at 300 K and 1 atm in J mol^{-1} is – xR, where x is _____. (Rounded off to the nearest integer)
[R = 8.31 J $mol^{-1}K^{-1}$ and ln 10 = 2.3]

Open solution

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Q130. JEE Main 2020 (Online) 5th September Evening Slot

For a reaction X + Y ⇌ 2Z , 1.0 mol of X, 1.5 mol of Y and 0.5 mol of Z were taken in a 1 L vessel and allowed to react. At equilibrium, the concentration of Z was 1.0 mol L^{-1}. The equilibrium constant of reaction is $\frac{x}{15}$. The value of x is _____.

Open solution

MCQ (More than One Correct Answer) 1 questions

Q131. JEE Main 2023 (Online) 1st February Evening Shift

The effect of addition of helium gas to the following reaction in equilibrium state, is :

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

(A) the equilibrium will shift in the forward direction and more of Cl_2 and PCl_3 gases will be produced.

(B) addition of helium will not affect the equilibrium.

(C) helium will deactivate PCl_5 and reaction will stop.

(D) the equilibrium will go backward due to suppression of dissociation of PCl_5.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **A**	3. **B**	4. **D**	5. **D**
6. **B**	7. **A**	8. **B**	9. **A**	10. **D**
11. **A**	12. **B**	13. **C**	14. **C**	15. **D**
16. **B**	17. **D**	18. **C**	19. **D**	20. **C**
21. **B**	22. **B**	23. **B**	24. **B**	25. **C**
26. **A**	27. **B**	28. **B**	29. **A**	30. **A**
31. **B**	32. **D**	33. **C**	34. **B**	35. **B**
36. **B**	37. **C**	38. **B**	39. **D**	40. **C**
41. **C**	42. **D**	43. **C**	44. **C**	45. **D**
46. **B**	47. **A**	48. **D**	49. **C**	50. **B**
51. **C**	52. **B**	53. **A**	54. **B**	55. **D**
56. **C**	57. **D**	58. **B**	59. **A**	60. **D**
61. **B**	62. **C**	63. **D**	64. **C**	65. **B**
66. **D**	67. **D**	68. **A**	69. **C**	70. **A**
71. **A**	72. **D**	73. **A**	74. **B**	75. **B**
76. **C**	77. **C**	78. **B**	79. **B**	80. **A**
81. **C**				

Numerical

82. **4**	83. **8**	84. **230**	85. **2474**	86. **15**
87. **125**	88. **33**	89. **5**	90. **74**	91. **962**
92. **417**	93. **2**	94. **1**	95. **44**	96. **3**
97. **3**	98. **49**	99. **12**	100. **1**	101. **3**
102. **4**	103. **2**	104. **6**	105. **2**	106. **2**
107. **29**	108. **125**	109. **2735**	110. **710**	111. **1107**
112. **747**	113. **6**	114. **4**	115. **182**	116. **0**
117. **16**	118. **1400**	119. **25**	120. **2**	121. **172**
122. **2**	123. **354**	124. **2**	125. **20**	126. **72 to 75**
127. **1**	128. **5**	129. **1380**	130. **16**	

MCQ (More than One Correct Answer)

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131. **A, B**

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